Ariel A. Arévalo Alvarado

AI Systems Architect & Researcher

arielarevaloalvarado@gmail.com ◇ (+506) 7064 1684 ◇ Heredia, Costa Rica ◇ Website ◇ LinkedIn ◇ GitHub

"Built 2 enterprise AI platforms from scratch as founding/primary engineer, delivering 50+ production agents and a product concept validated by a $9M investment from OpenAI in a similar initiative. Engaged in parallel practice in empirical AI evaluation and AI safety research."

Work History

Chief AI Architect (Founding) · **CRSS**, Remote — *Jun 2025 – Present*

- Established an AI engineering organization spanning 33 repositories and 10 engineers, including production platforms, client prototypes, documentation hubs, and team training resources.
- Designed an apprentice agent workflow platform (AutoActivity) from zero to production as the chief architect, delivering a 4-repository system spanning backend (FastAPI), frontend (Playwright browser event trace capture), and Azure ML data pipelines.
- Architected and shipped the TracePipe pivot, extracting AutoActivity's browser-trace capture, processing, and canonical-action classification into a standalone product after a competitor raised $9M from OpenAI for the same concept.
- Architected and built across 5+ client-specific AI prototypes (chatbots, document classifiers, and data analysis agents) using LangChain, MCP, Azure AI Foundry, and Slack integrations.

Skills: Python, FastAPI, Azure ML, LangChain, Docker, MCP, Alembic, Azure AI Foundry, System Architecture

Senior AI Engineer · **Colibri Group**, Remote — *Jul 2025 – Present*

- Initiated the Colibri AI Foundations enterprise strategy, autonomously building an in-house stack across infrastructure (model routing, key serving, telemetry), platform (*Instant Agents*), and tooling (harness, observability). Company leadership promoted it to a dedicated team and initiative, which I now lead.
- Designed a zero-deployment agent configuration architecture (*Instant Agents*) that reduced per-agent deployment effort by 45% through a control plane/data plane separation with immutable versioning and automated component discovery. Scaled from 6 to 50+ agents at $8.52/agent/month with under 1% infrastructure overhead.
- Built the Rubi enterprise AI platform from scratch as the primary developer, implementing agent runtime orchestration, JWT security, persistent memory, and a self-serve agent builder on AWS Bedrock AgentCore.

Skills: Python, AWS Bedrock AgentCore, MCP, Abacus AI, System Architecture

AI/ML Engineer · **Datasite**, Heredia — *Apr 2024 – Jun 2025*

- Designed the first end-to-end MLOps pipeline at Datasite, replacing ad-hoc Jupyter notebook experimentation with reproducible Azure ML pipeline steps incorporating data versioning and experiment tracking.
- Architected a medallion data architecture (Bronze/Silver/Gold) for production ML workflows, establishing standardized ingestion, validation, and feature engineering stages on Airflow and Azure ML.
- Maintained and extended existing production systems running classical NLP methods (NER, PII detection, BM25 search), regression models with Scikit-learn, and deep learning classifiers (MLPs) using PyTorch and Keras.

Skills: Python, Pandas, Azure ML, Apache Airflow, MLOps, Scikit-learn, Keras, PyTorch, NLP, Deep Learning

Data Engineer · **TransUnion**, Heredia — *Dec 2022 – Apr 2024*

- Led development of the initial ETL pipeline synthesizing IPv6 geolocation data for the industry-leading IP GeoPoint product, designing Spark SQL and Spark ML workflows on Google Cloud Dataproc.
- Maintained an ETL pipeline synthesizing IP reputation data to determine whether users were malicious and/or bots, making use of a Decision Tree model running in J2EE.

Skills: Scala, Java, Python, Apache Spark, Spring Framework, GCP Dataproc

Software Developer · **Snap Finance**, Alajuela *Dec 2021 – Dec 2022*

- Developed originations back-end services across monolith (J2EE), microservices (Spring), and serverless (AWS Lambda) architectures for consumer financing applications.

Skills: Java, JavaScript, Python, Spring Framework, J2EE, AWS Lambda

Projects

Moral Licensing Comprehension ↗: *Measuring Flexibility of Social Prediction in LLMs* *Apr 2026*

- Solo submission to the Google DeepMind/Kaggle *Measuring Progress Toward AGI* competition. Tested whether LLMs predict that past virtue *frees* rather than *binds* others' future behavior (120 items). Across 32 models, binary-choice accuracy ranged from 3% to 76%, and in most model families the larger models scored lower.

Skills: Python, Claude Agent SDK, OpenRouter, Pytest, Statistical Analysis, Empirical Evaluation

Rigger ↗: *Declarative coding-agent harness framework* *Mar 2026*

- Built and released as Apache-2.0 OSS on PyPI a Python framework for declarative coding-agent harness management, with users defining harnesses as code or configuration and the framework managing their development lifecycle from workflow specifications.

Skills: Python, Claude Agent SDK, Automated Software Engineering, Evolutionary Algorithms

ClimateFact ↗: *AI-powered climate statement fact-checker* *Spring 2025*

- Built an AI fact-checking system using LangGraph and Azure OpenAI to detect contradictions in climate statements against the IPCC AR6 report, leveraging DeBERTa natural language inference models for semantic validation.

Skills: Python, LangGraph, Azure OpenAI, NLP, NLI, DeBERTa

Publications

ACCEPTED *Funded by a BlueDot Impact Rapid Grant*

A. Arévalo-Alvarado and A. Badilla-Olivas, "Known-groups validity is not enough for evaluation-awareness detectors", *Agent Evaluation Science Symposium, Fall 2026* ↗.

- Ran five detectors (three text, two white-box) on 25,200 rollouts across two open-weight models, with public code and results.
- Found that every detector passes a known-groups check, that random activation directions pass it too, and that the activation probe agrees with no other detector.

PUBLISHED *Funded by CRSS*

K. Villalobos-Solís, E. Víquez-Mora, A. Badilla-Olivas, E. Vílchez-Lizano, and [†]**A. Arévalo-Alvarado**, "Multimodal Foundation Models for Zero-Shot User Activity Classification", *IEEE JoCICI 2026* ↗.

- Compared five multimodal LLMs on user-action classification from structured logs, screenshots, and both. Logs gave more consistent signals, and models were often confident when wrong.
- As senior author, conceptualized the study, supervised four student co-authors, and secured funding and compute.

[†]Senior author

Professional Development

Anthropic, Claude Certified Architect - Foundations	*Nov 2026*
BlueDot Impact, Technical AI Safety Project	*May – Jun 2026*
BlueDot Impact, Technical AI Safety	*Apr – May 2026*
BlueDot Impact, AGI Strategy	*Feb – Mar 2026*